

ESP32 RFID Anti-Wall-Jumping System

Advanced multi-sensor security solution for perimeter protection

Overview

The **ESP32 RFID Anti-Wall-Jumping System** is a robust, real-time perimeter security prototype designed to detect unauthorized access (like wall climbing or jumping) and respond instantly. It leverages multiple ultrasonic sensors for detection, an RFID reader for authorized access, a DFPlayer Mini for audio alarms, and a SIM800L module for instant SMS notifications to security personnel.

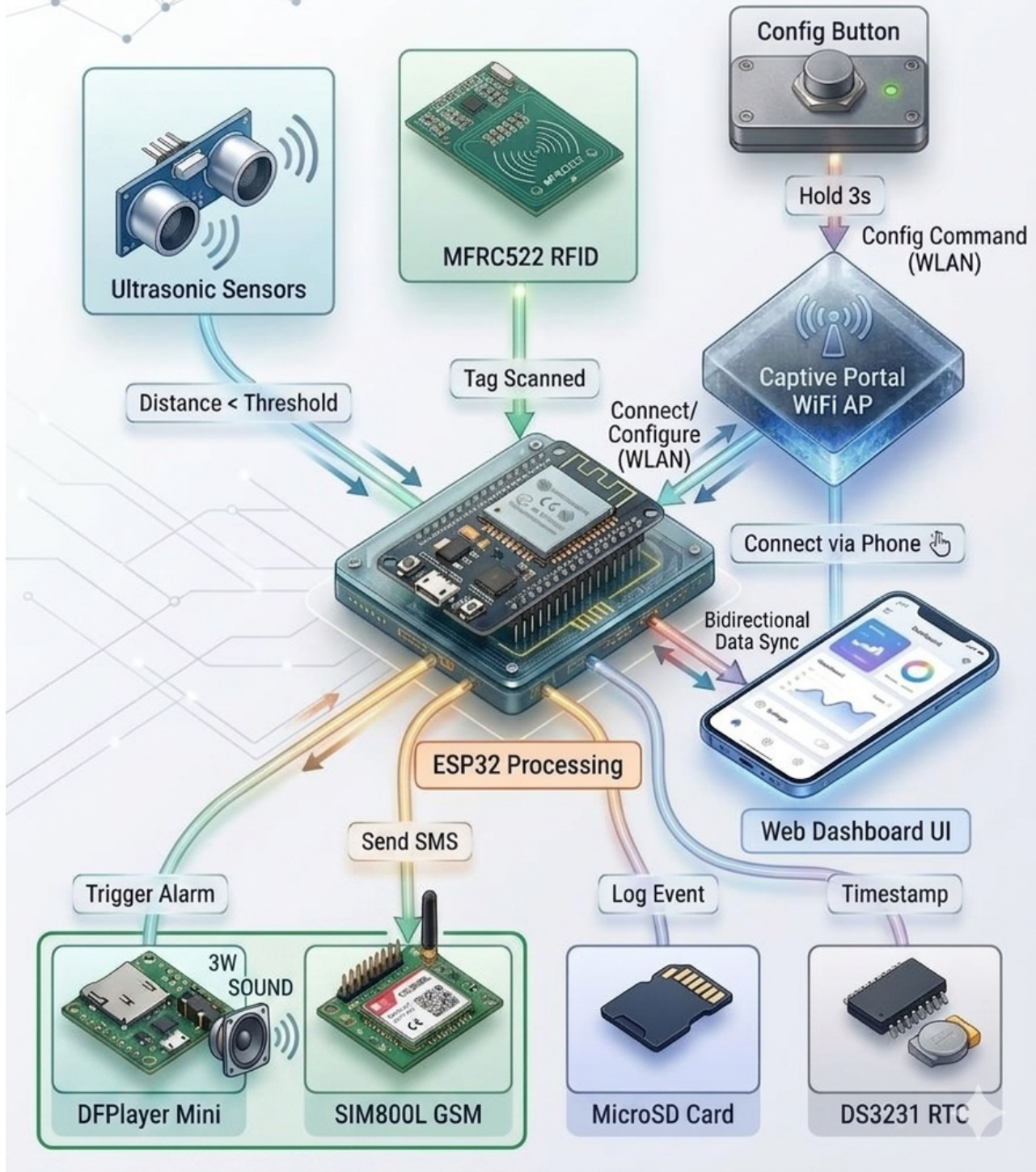
The system features a **built-in Captive Portal (Web UI)** for easy configuration on the fly, avoiding the need to re-flash the firmware to change settings.

Key Features

-  **Multi-Sensor Perimeter Detection:** Supports up to 3 separate HC-SR04 ultrasonic sensors dynamically configurable for wide coverage.
 -  **Instant SMS Alerts:** Uses SIM800L to send immediate notifications to a registered emergency contact upon intrusion.
 -  **Audio Alarms:** Integrates DFPlayer Mini to broadcast custom localized voice alarms or sirens.
 -  **RFID Integration (MFRC522):** Allows authorized personnel (e.g., guards or authorized students) to bypass or interact with the system.
 -  **Local SD Card Logging:** All events and access attempts are saved to an onboard MicroSD card in FAT32 format.
 -  **Web-Based Captive Portal:** Built-in async web server allowing administrators to configure system thresholds, modes, and sensors over Wi-Fi without compiling code.
 -  **Real-Time Clock:** Built-in DS3231 RTC for accurate time-stamping of intrusion logs.
 -  **Hardened Firmware:** Production-grade concurrency protection with dedicated FreeRTOS mutexes, deferred restart handling, validated RTC input, and guarded SD operations.
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System Architecture

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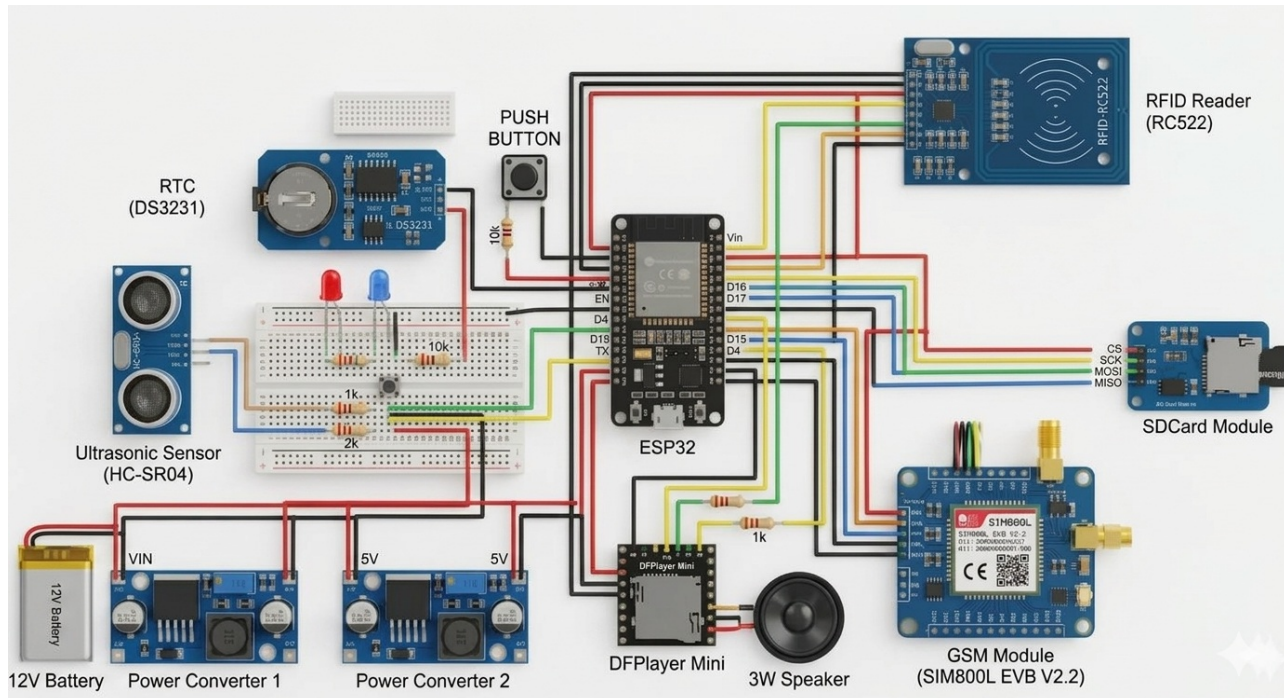


Hardware Stack

Component	Function
ESP32-38P	Core processing and Web Server AP
MFRC522	RFID Scanning

HC-SR04 (x3)	<i>Ultrasonic motion/distance detection</i>
SIM800L	<i>GSM Module for SMS alerts</i>
DFPlayer Mini	<i>MP3 Audio playback</i>
DS3231	<i>Precision RTC for logging</i>
MicroSD Module	<i>Local event storage</i>

Realistic Component Layout



[!NOTE]

Visual Reference Only: The realistic component layout above is for visual purposes and aesthetic representation. For exact pin mapping and electrical connections, strictly follow the official 2D Schematic Circuit Diagram below.

Schematic Circuit Diagram

For full hardware wiring instructions and component mapping, please refer to the official wiring schematic:

To build this project, we need to connect several parts to our main brain, the **ESP32 microcontroller**. Make sure all the wires are connected securely before turning on the power.

1.1 Power Connections

First, let's connect the power! Different parts need different amounts of power to work safely.

[!CAUTION]

Important Power Rule: The SIM800L (our text message sender) needs a strong power source. Always power it from your **5V Buck Converter**, never directly from the ESP32!

Part	Power Pin (VCC)	Ground Pin (GND)	Where does it get power from?
ESP32 (Main Brain)	5V (Vin)	GND	5V Buck Converter
SIM800L (Text Module)	5V	GND	5V Buck Converter
DFPlayer Mini (Voice Player)	VCC	GND	5V Buck Converter
HC-SR04 (Distance Sensors)	VCC	GND	5V Buck Converter
SD Card Reader	VCC	GND	5V Buck Converter
RFID-RC522 (Card Reader)	3.3V	GND	ESP32's 3V3 pin
RTC-DS3231 (Clock)	VCC	GND	ESP32's 3V3 pin

1.2 Pin Connections

Now, let's connect the data pins so the ESP32 can talk to the parts. Follow this table carefully to wire everything up.

Part	Pin Name	Connect to ESP32 Pin
Mode Button	Button Pin	GPIO 34
Sensor #1 (HC-SR04)	TRIG	GPIO 14
	ECHO	GPIO 12
Sensor #2 (HC-SR04)	TRIG	GPIO 33
	ECHO	GPIO 25
Sensor #3 (HC-SR04)	TRIG	GPIO 32
	ECHO	GPIO 35
Clock (RTC-DS3231)	SCL	GPIO 26
	SDA	GPIO 27
Blue LED (Status)	Positive Leg	GPIO 13
Red LED (Alarm)	Positive Leg	GPIO 4
Card Reader (RFID)	MOSI	GPIO 23
	RST	GPIO 22
	SS (or SDA)	GPIO 21
	MISO	GPIO 19
	SCK	GPIO 18
SD Card Reader	CS	GPIO 5
	MOSI	GPIO 23 (Shared with RFID)
	MISO	GPIO 19 (Shared with RFID)
	SCK	GPIO 18 (Shared with RFID)
Text Module (SIM800L)	TX	GPIO 17
	RX	GPIO 16
Voice Player (DFPlayer)	TX	GPIO 2
	RX	GPIO 15

1.3 Setting Up the Voice Player

We use a MicroSD card to store the alarm sounds for the DFPlayer Mini.

1. Get a MicroSD card and format it to **FAT32**.
2. Create a folder named exactly **mp3** on the card.
3. Add your sound files inside the folder and name them exactly like this:
 - 0001.mp3 (e.g., "Please return to class...")
 - 0002.mp3 (e.g., "Security alert...")
 - ...and so on.
4. Put the card into the DFPlayer Mini module.

2. How the System Works

The system has two main states or "modes". You can switch between them by holding down the **Mode Button**.

2.1 Normal Mode (Guarding)

- **Blue Light:** Stays ON.
- **What it does:** The system uses the distance sensors to watch the wall. If someone comes too close (meaning they are trying to jump the wall!), it triggers the alarm.
- **When triggered:** It plays a warning sound, flashes the red light, and sends a text message to a security guard.

2.2 Setup Mode

- **Blue Light:** Blinking fast.
- **How to enter:** Press and hold the **Mode Button** for 3 seconds.
- **What it does:** The system stops watching the wall and creates its own Wi-Fi network. You can connect to this Wi-Fi to change settings using your phone or laptop.
- **How to exit:** Press and hold the button for 3 seconds again. (It will also exit automatically after 1 hour if you forget).

2.3 Remote Control via Text Message (SMS)

You don't always have to press the physical button to change modes! You can send a text message to the system's SIM card to control it remotely:

- **Text AP ON:** Turns on the Wi-Fi Setup Mode so you can connect to the dashboard.
- **Text AP OFF:** Turns off the Wi-Fi Setup Mode and goes back to Normal Guarding mode.

(Note: For extra security, if "Number Lock" is enabled, the system will only obey text commands sent from the registered security guard's phone number!)

3. Changing Settings on the Dashboard

3.1 Connecting to the Setup Page

1. Put the system into **Setup Mode** (hold the button until the blue light blinks).
2. On your phone or laptop, connect to the Wi-Fi network:
 - **Wi-Fi Name:** AntiWallJump-Config
 - **Password:** configure123
3. A login page should pop up. If it doesn't, open a web browser and go to `http://192.168.4.1`.
4. The default admin password is `admin`.

3.2 What's on the Dashboard?

The setup page has different sections and buttons to help you control the system:

Top Menu Buttons (Quick Actions):

- **Sync Clock (Clock Icon):** Click this to instantly set the system's time to match your phone or computer.
- **Dark/Light Mode (Moon/Sun Icon):** Switch the screen between dark and light colors.
- **Logout (Door Icon):** Securely sign out of the setup page.
- **Stop Portal (Power Icon):** Quickly exit setup mode and turn the alarm back on right away.

Bottom Navigation Tabs:

- **Dashboard:** See the "Node Overview" (system health like the memory card status) and check the overall security status.
 - **Registry:** Register student ID cards and see who triggered the alarm.
 - **Settings:** Change the phone number for alerts, choose how close someone has to be to trigger the alarm, and pick which sound plays.
 - **Records:** View a history of all past alarms and download a record file.
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4. Student ID Cards (Registry)

The system keeps track of students using their ID cards.

4.1 Adding a New Student Card

1. Go to the **Registry** tab on the dashboard.
2. Click **Scan Card**.
3. Tap a new ID card on the RFID reader. The card's special ID will show up on the screen.
4. Type in the student's name and school ID number.
5. Click **Register** to save them in the system.

4.2 What happens when the alarm triggers?

- If the sensors spot someone climbing, it logs an **UNKNOWN INTRUDER**.
 - If someone taps a registered student ID card on the reader, the alarm triggers and records that student's name, so the school knows who it was.
 - If an unregistered card is tapped, it also triggers the alarm to keep the area secure.
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5. Alarm Records

The system saves a history of every time the alarm goes off. It saves this inside its internal memory and on a MicroSD card, so nothing gets lost!

5.1 Downloading the Records

You can download a spreadsheet file of all the past alarms:

1. Go to the **Records** tab on the dashboard.
2. Click **"Download Logs"**.
3. A file called `Campus_Security_Logs.csv` will download to your device. You can open this with Excel or Google Sheets to see a neat list of everything that happened.